Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech (EE) Examination

May 2016

EE307 Electrical Power System III

Date: 19/05/2016, Thursday **Time: 10.00 a.m. to 1.00 p.m** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Fill in the blanks with appropriate word or words. **[07]**

I. Power factor of an arc is _______.
II. As the natural frequency of oscillation increases, the breaking capacity of the CB ________.
III. The function of magnetic blowout coil is __________.
IV. The RRRV becomes zero at _______.
V. Making current of circuit breaker is _______ than the breaking current of that circuit breaker.
VI. For 415 Volts application _________ CB is used.
VII. The main difference between CB and switch is _______.

Q – 2. A In connection with circuit breaker operation, with neat diagrams and suitable example, explain the phenomena of current chopping. **[06]**

B With neat diagram explain the construction and working of Vacuum circuit breaker. **[08]**

OR

Q – 2. A Explain the difference causes due to which deionization of gases take place. **[05]**

B List out the factors that affect the TRV of a circuit breaker. Explain any three of them. **[04]**

C Write techinical detail about rating of AC circuit breaker. **[05]**

Q - 3 **Answer any TWO**

A In a short-circuit test, with isolated neutral on a 220 kV, 3-phase circuit breaker, the power factor of the fault was 0.3, the recovery voltage was 0.88 of full-line value, the breaking current was symmetrical and the restriking transient had a natural frequency of 22,000 Hz. Estimate the average rate of rise of restriking voltage and $RRRV_{Max}$. **[07]**

B Explain the principle of resistance switching. Why is it necessary in air blast C.B.? Derive the equation of critical damping resistance. **[07]**

C With neat diagram explain the construction and working of Air Blast Circuit Breaker. **[07]**

SECTION – II

Q - 4 Choose the correct option. **[07]**

I. If negative sequence current component is zero in a fault current, the possible fault is…

(a) L-G (b) L-L (c) L-L-G (d) L-L-L

II. If the positive-, negative- and zero-sequence reactances of an element of a power system are 0.3, 0.3 and 0.08 pu respectively, then the element would be a

(a) synchronous generator (b) transformer
(c) static load. (d) transmission line.

III. For the L-G fault at shown location of figure 1, as value of Xn increases, the effect on fault current is…..

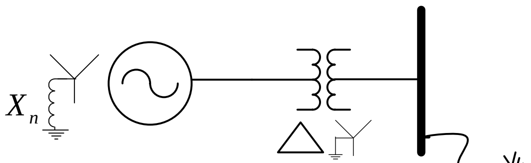
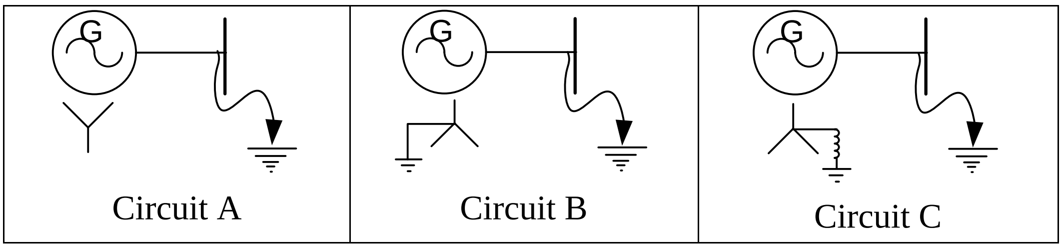


Figure 1

(a) Fault current increases (b) Fault current decrease
(c) Fault current remains same (d) No fault current will flow

IV. Which circuit has the maximum fault current for L-L fault at generator terminal?



(a) Circuit A (b) Circuit B (c) Circuit C
(d) All the circuit has same L-L fault current.

V. For the R_{phase} to Ground fault shown in figure 2, the fault current is 3 pu. The positive sequence current flowing from generator to transformer in R phase is..

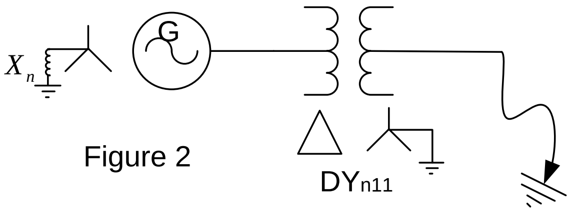


Figure 2

(a) 3pu (b) 1.73 pu (c) 1.0 pu (d) zero

VI. For the system shown in figure 3, if the pu voltage at bus B is 1 pu and the load at bus B are as shown in figure, The voltage at bus A is..

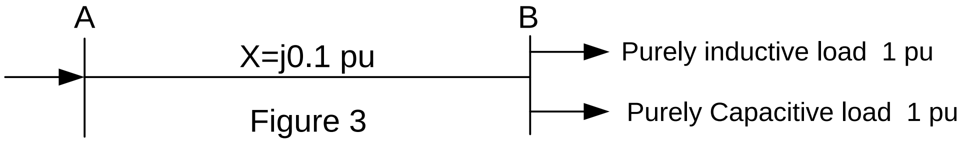


Figure 3

(a) 1.2 pu (b) 1.02 pu (c) 1 pu (d) Given data are insufficient.

VII. The sequence components of fault current are as under.

$I_{a1} = 2.3\angle 20^0 \ \ pu; \ I_{b2} = 2.3\angle 140^0 \ \ pu; \ I_{c0} = unknown\,value$

The possible fault is…

(a) L-G (b) L-L (c) L-L-G (d) given data are insufficient.

Q – 5. A Draw the zero sequence network diagram of figure 4. (**Take Base as per generator 1 rating**) **[07]**

The ratings and zero sequence reactance of equipments are as per following table. Reactance of line 1 and line 2 are 100 Ω each and that of line 3 is 10 Ω.

Equipment	Rating (MVA)	Voltage (kV)	Zero sequence reactance (%)
Generator 1	50	11	5
Generator 2	50	6.6	5
Motor 1	40	10	8
Motor 2	30	15	8
Transformer 1	50	11/132	10
Transformer 2	40	6/15	12
Transformer 3	45	132/11	12
Transformer 4	25	132/15	15

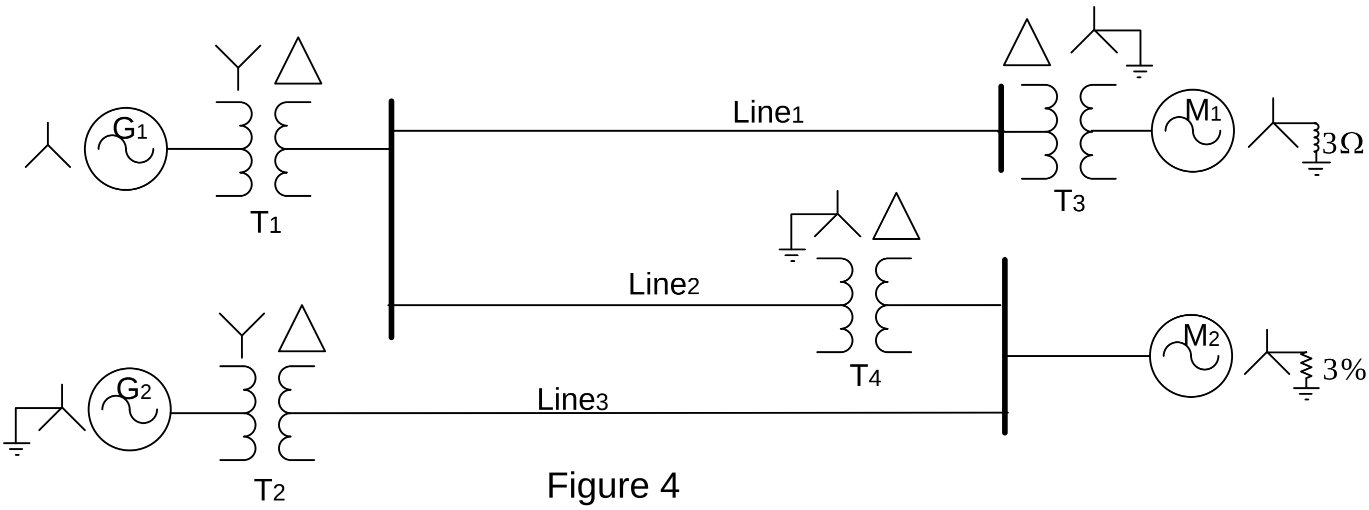


Figure 4

B Figure 5 shows a part of power system. The ratings of the equipments are as per figure. **[07]**

If the motor is a synchronous machine, drawing 75% of full laod current with the terminal voltage 10.5 kV and power factor 0.8 leading. At this time three phase fault occurs on the High voltage terminal of transformer T_1. Find the total fault current. (**Take base as per Generator rating.**)

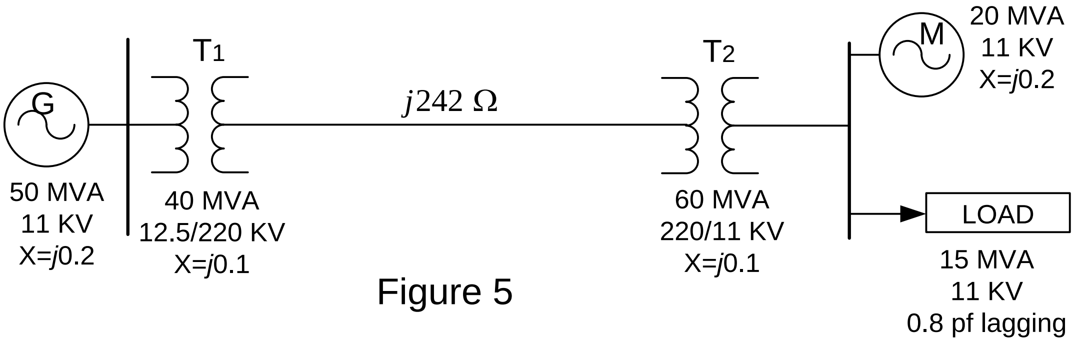


Figure 5

OR

Q – 5. A A sample power system shown in figure 6. The rating of generators and transformers are as per figure. The rectance of transmission lines are as per following table. **[07]**

Transmission line number	Reactance (Ω)
L1 and L3	87.846
L2 and L4	43.923
L5	131.769

Find out the 3 phase fault current magnitude when fault occurs at location'F' as per figure. (**Take base as per generator 1 rating.)**

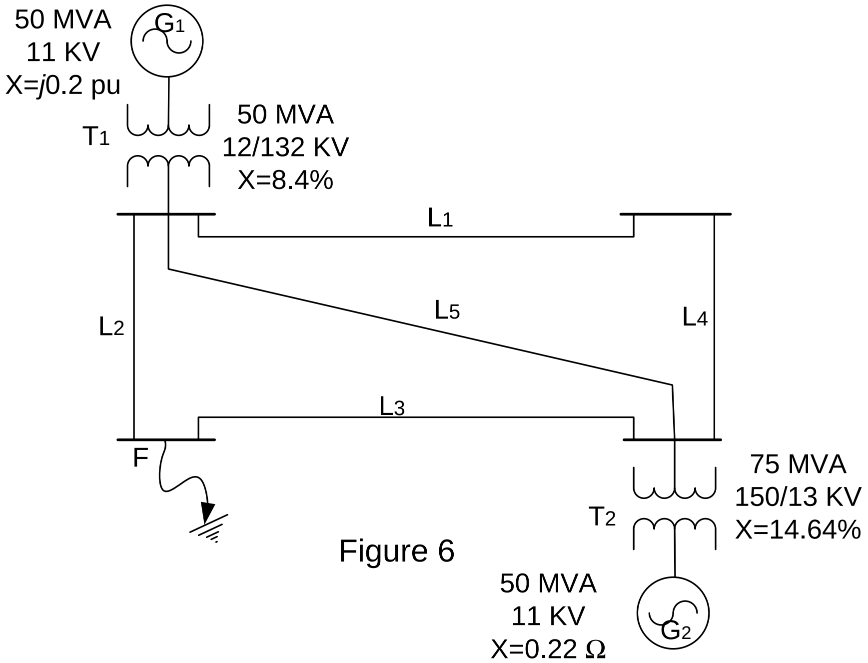


Figure 6

B Prove the power invariance theorem in connection with symmetrical component. **[07]**

Q – 6. Attempt any **Two**.

A Using appropriate interconnection of sequence networks, derive the equation for a line to line fault in a power system with a fault impedance of Z_f. **[07]**

B A part of power system is shown in figure 7. The ratings of equipments are indicated in figure. When all the motors are running at full load with terminal voltage of 6 kV, at that time **single line to ground** fault occurs at 'F' with zero fault impedance as shown in figure. **[07]**

Find out the total fault current and current passing through CB 'A' during fault. **(Take base as per generator Rating.)**

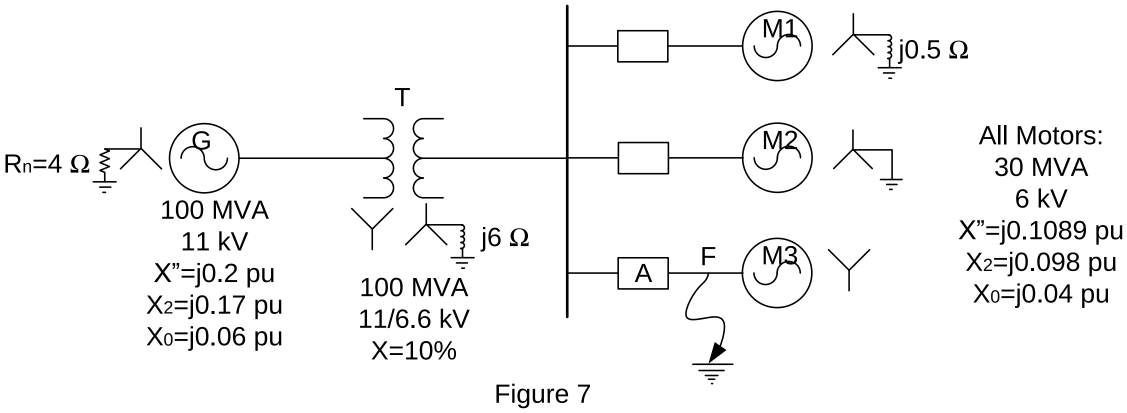


Figure 7

C A part of power system is shown in figure 8. The ratings of equipments are indicated in figure. The positive sequence reactance of each transmission line is 36.3 Ω, and zero sequence reactance of each line is 108.9 Ω. If double line to ground fault with zero fault resistance occurs at bus A, find out the magnitude of total fault current. **(Take base as per generator Rating.)** **[07]**

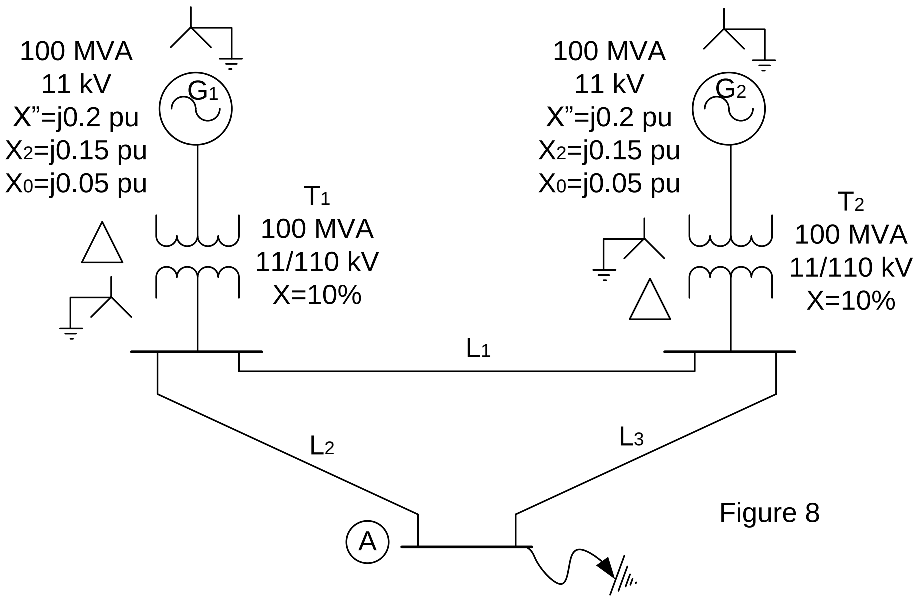


Figure 8

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